

Statistical Analysis: Plot Explanations

This document provides a detailed explanation of the statistical plots found in the *Statistical_Plots_Appendix.pdf*, covering hormones 1 through 8 and hormone 10_generated, as well as the correlation heatmaps.

1. Distribution Plots (Histograms with KDE)

For each hormone (e.g., hormone1 through hormone8), the first plot is a **Histogram** overlaid with a **Kernel Density Estimate (KDE)** curve.

- **X-Axis:** Represents the measured value of the specific hormone.
- **Y-Axis (Count):** Represents the frequency or number of occurrences of those values in the dataset.
- **Color Coding (Outcome 0 vs 1):** The blue/grey bars (Outcome 0) and green bars (Outcome 1) allow for a visual comparison of how hormone levels differ between the two outcome groups.

General Observation: Most hormones in this dataset (especially hormone1, hormone2, hormone3, and hormone5) exhibit **positive skewness** (right-skewed), meaning most data points are clustered at lower values with a long tail extending toward higher values.

2. Q-Q Plots (Quantile-Quantile Plots)

The second plot for each hormone is a **Normal Q-Q Plot**, used to assess if the hormone data follows a normal distribution.

- **Theoretical Quantiles:** What we would expect if the data were perfectly normally distributed.
- **Sample Quantiles:** The actual observed distribution of the data.

- **The Red Line:** Represents the "perfect" normal distribution.

General Observation: In almost all cases (e.g., Page 1, 2, 3, 5), the blue dots curve significantly away from the red line at the ends. This confirms that the hormones are **not normally distributed** and have "heavy tails" or outliers, particularly on the higher end.

3. Spearman Heatmaps (Correlation Analysis)

Pages 10 and 11 contain **Spearman Rank Correlation Heatmaps** for Outcome 0 and Outcome 1, respectively.

- **Purpose:** To measure the strength and direction of the monotonic relationship between pairs of hormones.
- **Scale (1.0 to -1.0):** A value of 1.0 (dark red) indicates a perfect positive correlation, while values near 0 (grey/light blue) indicate no relationship.
- **Key Finding:** There is a strong positive correlation between **hormone3** and **hormone4** (0.59 in Outcome 0 and 0.88 in Outcome 1), suggesting these two variables move together closely.

4. Hormone-Specific Notes

- **Hormone 10_generated (Page 9):** This plot shows a very extreme distribution where almost all values are near zero, with a single extreme outlier near 250, causing a near-vertical line in the Q-Q plot.
- **Hormone 8 (Page 8):** Compared to the others, hormone8 appears more spread out and slightly closer to a bell shape, though it still maintains a rightward skew.
- **Outcome Comparison:** In the distribution plots, Outcome 1 (green) typically has a much lower count than Outcome 0 (grey), suggesting the dataset is imbalanced.